



REDEFINING
MARITIME
SUPERIORITY_

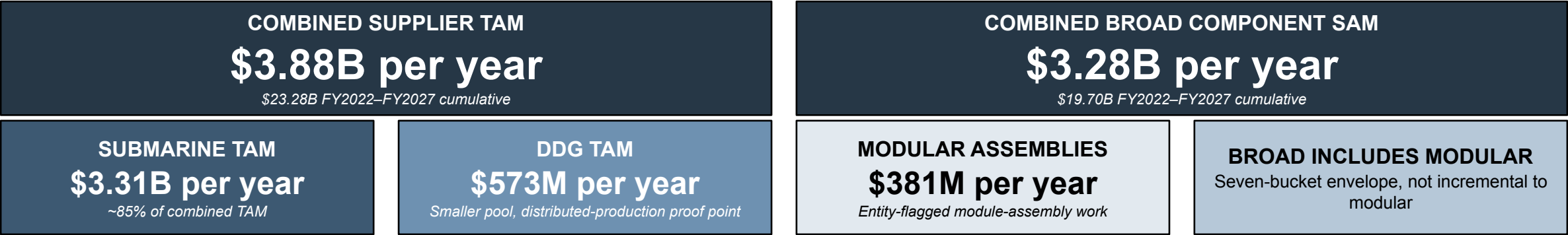
Distributed Shipbuilding Supplier Market

DDG and submarine new-construction component suppliers, FY2022–FY2027

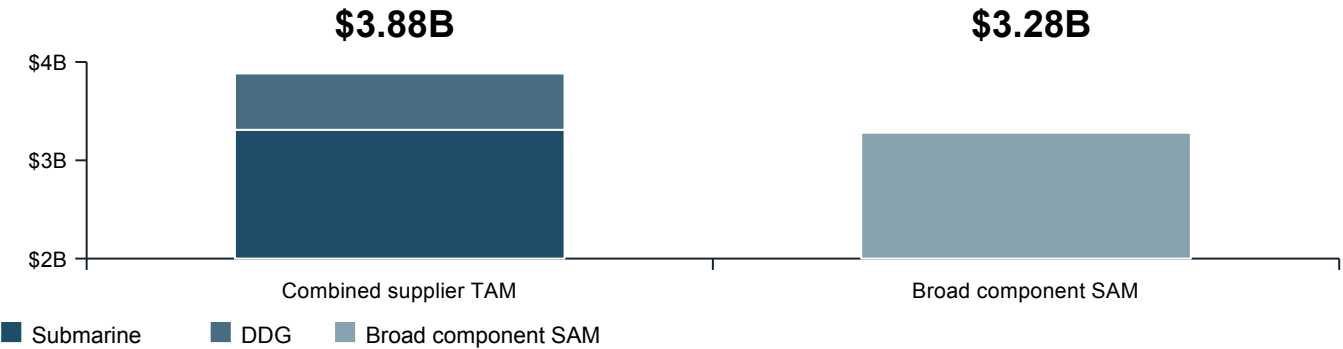
June 2026

Supplier TAM and SAM | Combined supplier opportunity is ~\$3.9B per year, with ~\$3.3B addressable as broad component SAM.

FY2022–FY2027 model-period averages, annualized; nominal then-year dollars.



Combined supplier TAM stacked by program vs broad component SAM, \$B per year; FY2022–FY2027 average, nominal then-year.



The pool is submarine-led, while DDG supplies the distributed-production proof point.

- Virginia/Columbia contributes \$3.31B of the \$3.88B annual TAM; DDG adds \$573M per year and carries the clearest public distributed-production signal.

Broad SAM is the serviceable envelope, not a capture forecast.

- Broad component manufacturing is \$3.28B per year across the seven named buckets; modular assemblies is a narrower \$381M entity-flagged cut inside that envelope.

The answer is a submarine-led supplier pool near \$3.9B per year, with DDG as the distributed-production proof point.

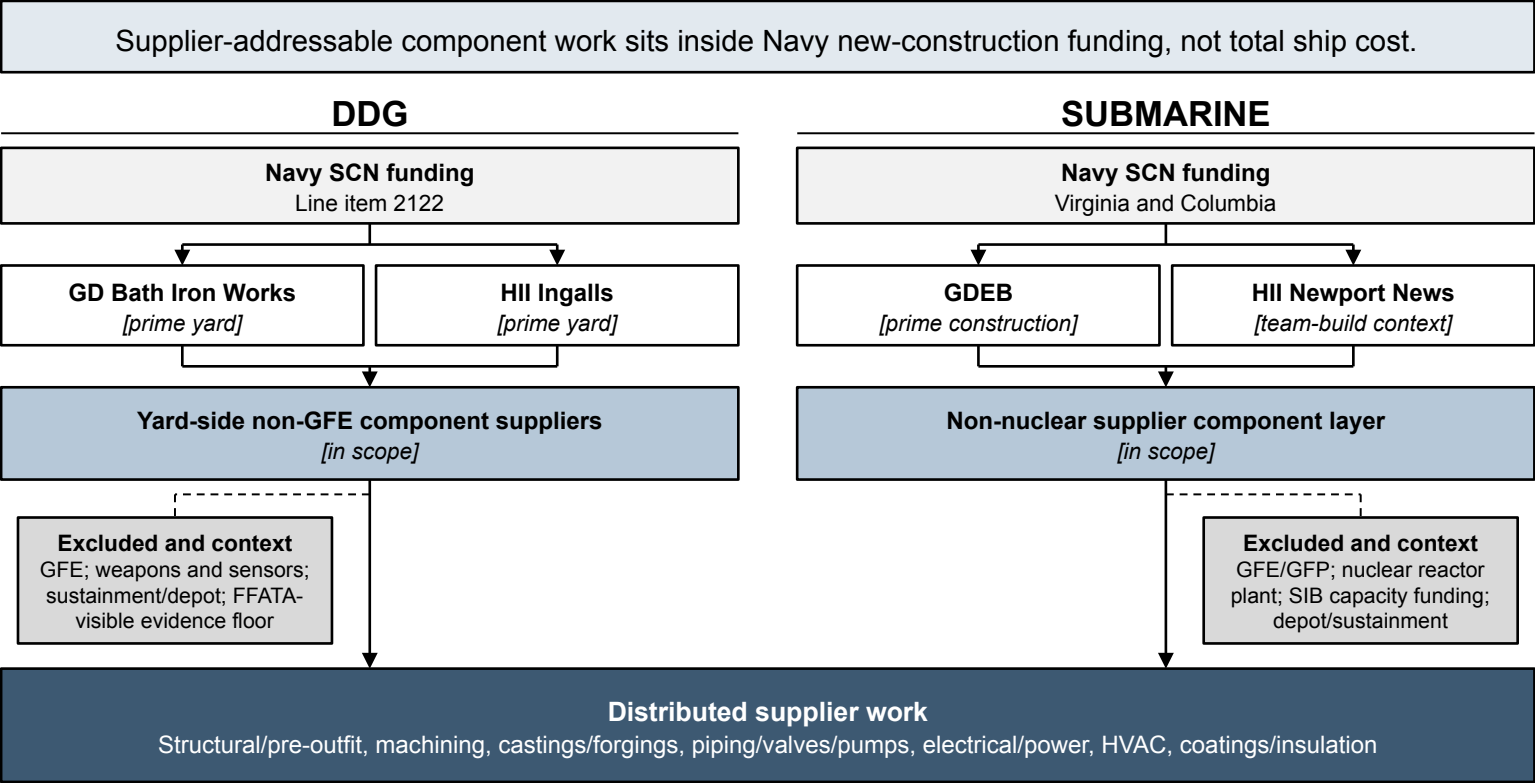
Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DoD place-of-performance award corpus and parent-prime PIID scope; (3) FFATA/FSRS subaward records and the operating-entity supplier registry

Market Landscape

The opportunity is the component layer yards buy out, and demand is funded



Supplier Layer | The opportunity sits between Navy budget authority and prime-yard construction, in component work yards buy out rather than self-perform.



The primes differ, but the serviceable cut is the same.

- DDG dollars run through Bath Iron Works and Ingalls; submarine dollars run through Electric Boat and Newport News. In both lanes, the counted layer is supplier component work outside GFE, sustainment, and yard self-perform.

Visible subawards prove the layer, not its full size.

- FFATA-visible filings provide supplier names and work-type cues; first-tier visibility thins below the first tier and differs by yard.

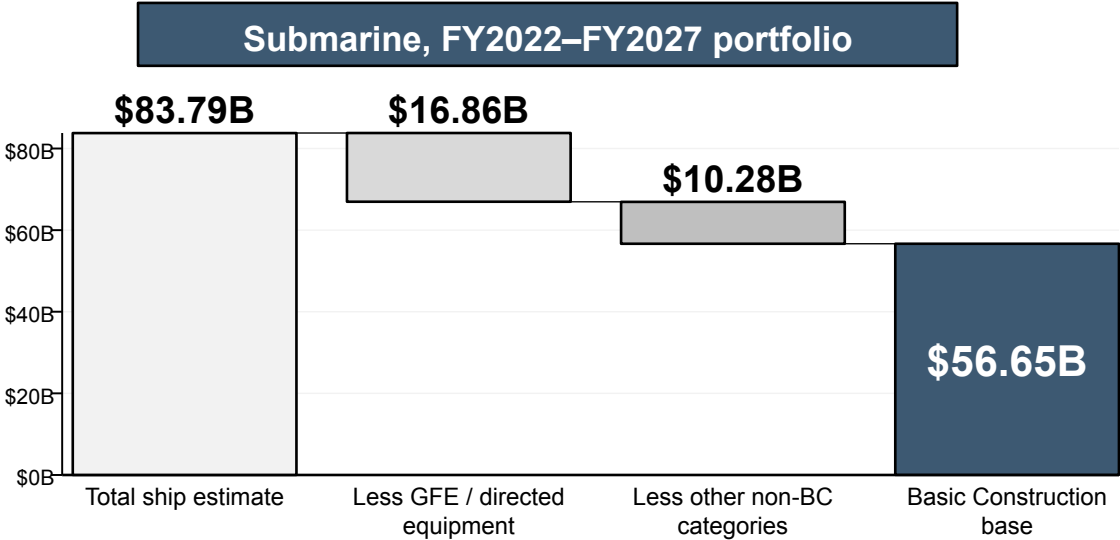
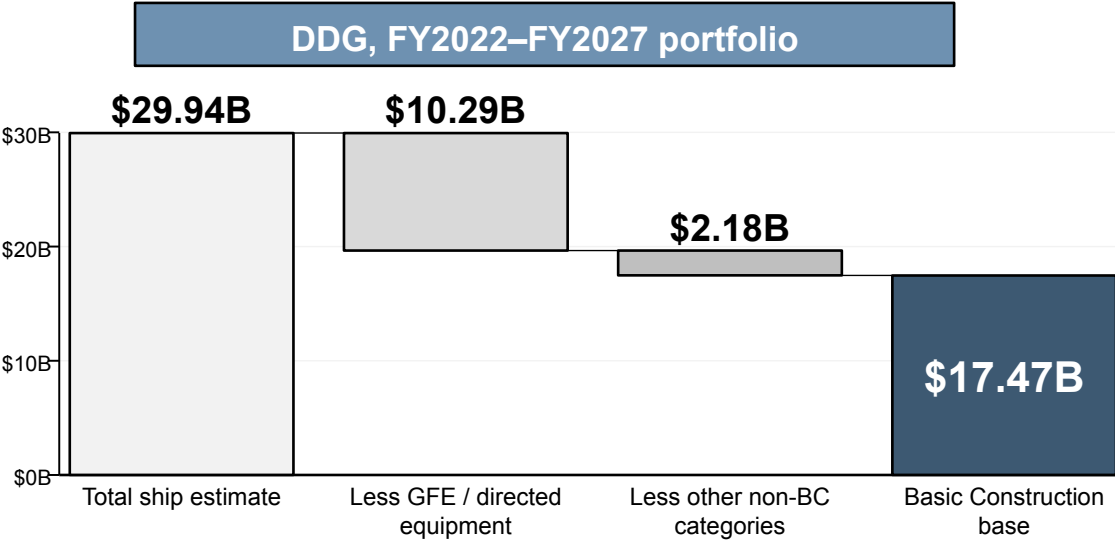
Electrical and power means ship power, distribution, and generation; combat and mission electronics leave the boundary before bucket assignment.

Blue = counted supplier layer; gray = excluded or context only; outline = prime or team-build layer.

Sources: (1) Navy SCN shipbuilding budget justification and line-item structure, FY2022–FY2027; (2) prime-yard place-of-performance award corpus; (3) FFATA/FSRS first-tier subaward records

Scope and Cost Funnel | Sizing starts at new-construction budget lines and narrows to non-GFE Basic Construction, not total ship cost.

FY2022–FY2027 cumulative, nominal then-year \$B. Each funnel uses its own vertical scale and narrows total ship estimate to the non-GFE Basic Construction base.



Basic Construction is the included denominator.

- The sizing base is \$17.47B of DDG's \$29.94B FY2022–FY2027 estimate and \$56.65B of the submarine portfolio's \$83.79B. This base funds yard labor, team-build work, purchased material, and first- and lower-tier supplier flow.

GFE, other budget categories, and yard self-perform leave the boundary.

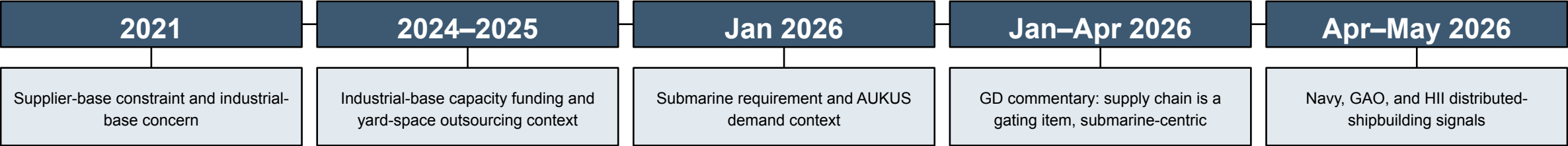
- GFE leaves first — DDG electronics and ordnance (\$10.29B); submarine government-furnished propulsion, electronics, and ordnance (\$16.86B). Then other non-BC categories — plans, change orders, HM&E, and other, plus Virginia technology insertion on the submarine side (\$2.18B DDG, \$10.28B sub). Yard self-perform within Basic Construction is removed later, at the supplier step.

Top-of-funnel context and first-tier filings are evidence, not numerator.

- Total ship estimate and TOA frame the denominator; FFATA/FSRS subawards are a visible floor behind the supplier coefficient. Do not add AP/LLTM on top of the BC base inside this funnel; the DDG-only filtered AP/LLTM stream is handled separately.

Sources: (1) DDG and submarine SCN/P-5c cost-funnel data, FY2022–FY2027; (2) Navy P-5c GFE and Basic Construction breakouts; (3) program TAM bridge inputs

Demand Backdrop | Policy, prime behavior, and industrial-base pressure point toward more distributed supplier capacity.



Supplier-base constraint: capacity, long-lead items, sole-source exposure, and industrial-base atrophy.

Policy and funding tailwind: Navy distributed-production direction and capacity investment.

Prime behavior: HII public distributed-work commitments; GD supply-chain pressure signals.

Demand signals support more external capacity, but not a larger sizing numerator.

- Policy, oversight, and prime commentary point toward more distributed supplier work; the FY2022–FY2027 model still sizes from SCN Basic Construction and strict supplier coefficients.

DDG and submarine signals support the same direction for different reasons.

- HII carries the clearer DDG distributed-production signal; GD's public signal is more submarine-centric, with margin pressure consistent with an outsourcing premium.



Virginia-class submarine under construction

The model sizes the current opportunity; future distributed work for new entrants is a diligence question, not a modeled input.

Sources: (1) Navy and OSD shipbuilding and industrial-base policy and budget materials, FY2021–FY2026; (2) GAO shipbuilding and submarine industrial-base reports; (3) HII and General Dynamics public disclosures and earnings commentary, 2024–2026; (4) submarine construction photograph, illustrative



Sizing the Opportunity

A budget-led TAM with an allocation-led SAM, submarine-led and lumpy by year



Methodology | TAM is budget-led, while SAM mix is allocated through one entity-resolved supplier registry.

Sizing chain and governing rules

LAYER	GOVERNING SIGNAL	TREATMENT	WHY IT MATTERS
Budget TAM	SCN Basic Construction and AP/LLTM stream	Build exogenous TAM by program	Classification does not move headline TAM
Supplier share	POP corpus	Strip prime, co-prime, GFE, and excluded scope	Converts the budget base into supplier opportunity
Supplier mix	Operating entity, SAM-resolved supplier registry	Role first, then bucket; subawards are mix evidence, not market size	One registry classifies DDG and submarine alike
Physical buckets	Seven stable work-type buckets	Structural, machining, castings, piping, electrical power, HVAC, coatings	Keeps the two programs comparable
Exclusions and residual	Mission systems, service, holding, foreign/FMS, unresolved tail	Route out, or hold residual explicit	Broad SAM does not auto-equal TAM
Scenario SAM	Bucket and entity flags	Overlapping cuts: metal, HM&E, electrical, modular, broad	Scenarios are not additive

- TAM is fixed by the budget method, not by supplier reclassification.**
- Supplier TAM is the budget base multiplied by the place-of-performance supplier share; re-bucketing suppliers changes mix, not the \$3.88B headline.
- The same registry governs both programs after role exclusions.**
- DDG and submarine dollars are classified at the operating-entity level after mission systems, services, holding companies, foreign/FMS, primes, and GFE are removed.
- Residual is a modeled result, not a cleanup item.**
- Unresolved supplier dollars stay in the denominator and dilute bucket shares, so broad SAM does not automatically equal total TAM.

Modeling boundary: electrical means ship power, distribution, and generation; combat and mission electronics are excluded in the base case. DDG VLS launch-control is sensitivity, not base scope.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DoD place-of-performance (POP) award corpus and parent-prime PIID scope; (3) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026



TAM Bridge | Combined supplier TAM is \$3.88B per year, but AP/LLTM is additive for DDG and reference-only for submarine.

Supplier TAM bridge, \$B cumulative and \$M per year; nominal then-year. Each program is bridged separately, then summed.

DDG supplier TAM bridge

BC construction base
\$17.47B cumulative
Less prime, co-prime, and GFE place-of-performance
\$15.28B removed scope removal, not a loss
BC supplier stream
\$2.19B cumulative \$365M per year
Add AP/LLTM stream
\$1.25B cumulative \$208M per year
DDG supplier TAM
\$3.44B cumulative \$573M per year

Submarine supplier TAM bridge

Basic Construction base
\$56.65B cumulative
Applied supplier share
35.0% strict, non-nuclear, yard-excluded
Submarine supplier TAM
\$19.84B cumulative \$3.31B per year
Additive AP/LLTM base
\$0 supplier long-lead already inside Basic Construction

↓

↓

COMBINED SUPPLIER TAM
\$3.88B per year \$23.28B cumulative

DDG has two additive supplier streams.

- Basic Construction supplier work is \$365M per year and filtered AP/LLTM adds \$208M per year, because DDG Basic Construction is net of prior-year AP.

Submarine AP/LLTM stays reference-only.

- The 35.0% supplier share on \$56.65B of Basic Construction yields \$3.31B per year; supplier long-lead material already sits inside the boundary, so additive AP/LLTM remains \$0.

The combined TAM sums dollars, not rules.

- DDG and submarine outputs sum to \$3.88B per year and \$23.28B cumulative; the AP/LLTM treatment remains program-specific and should not be blended.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DDG MYP-corrected outside-yards place-of-performance evidence; (3) submarine coefficient and AP/LLTM boundary analysis

Supplier Share | Strict inputs hold TAM conservative, but DDG and submarine solve different evidence problems.

DDG APPLIED BC SUPPLIER SHARE

12.5%

MYP-corrected non-GFE BC stream

SUBMARINE APPLIED BC SUPPLIER SHARE

35.0%

Strict non-nuclear Basic Construction view

PROGRAM	EVIDENCE ISSUE	APPLIED VALUE	BROADER EVIDENCE VIEWS	FEEDS HEADLINE TAM
DDG	MYP redaction	12.5% BC supplier share	Restored \$14.58B multiyear masters; corrected outside-yards POP ~32.8%	12.5% on the non-GFE BC stream
	AP/LLTM	85% supplier-share input on 80% ship-construction share	Assumption-driven filtered stream	Additive filtered stream, \$208M per year
Submarine	POP breadth	35.0% BC supplier share	All-gated POP anchor ~51.8%; AP/LLTM reference ~48.5%	35.0% on non-nuclear BC
	AP/LLTM	\$0 additive base	Gross AP is large supplier evidence	Not additive under the current boundary

DDG supplier-share evidence starts with a redaction correction.

- The model restores \$14.58B of redacted multiyear masters to correct the outside-yards POP read; the applied BC supplier share remains 12.5%.

Submarine supplier-share evidence is broad but deliberately constrained.

- The 35.0% input is a strict non-nuclear, yard-excluded view; broader POP anchors near 51.8% and 48.5% stay sensitivity.

Only the applied shares feed headline TAM.

- Broader POP views support the audit record, but 12.5% for DDG and 35.0% for submarine are the TAM bridge inputs.

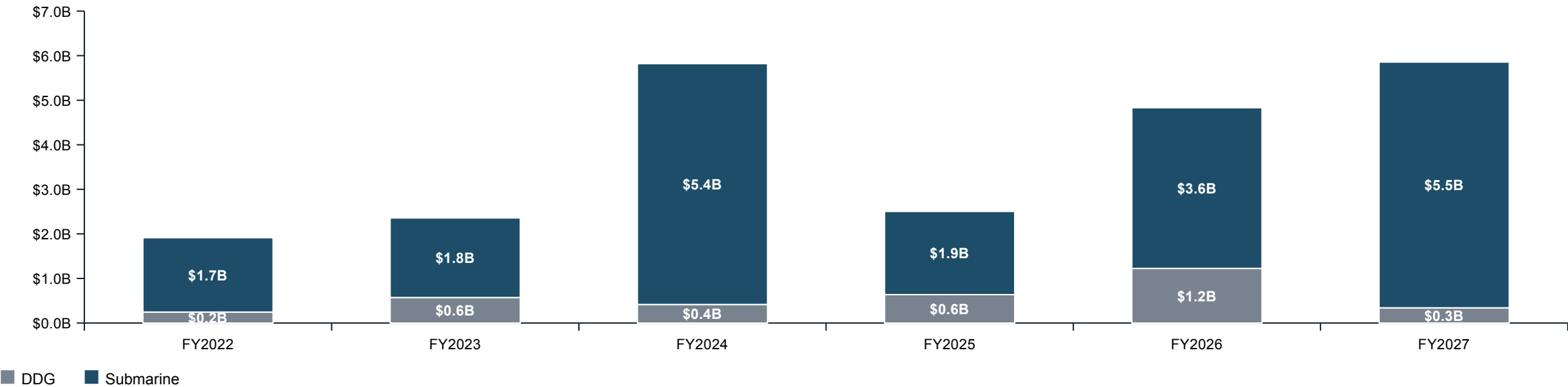
The two programs' shares are inputs to separate models; a single blended supplier-share percentage would mislead.

Sources: (1) DDG MYP redaction correction and outside-yards place-of-performance evidence; (2) submarine coefficient and AP/LLTM boundary evidence; (3) Navy SCN and P-5c Basic Construction budget justification, FY2022–FY2027



Annual Cadence | Average annual TAM is a sizing convention; actual supplier demand is lumpy and peaks in submarine-driven FY2024 and FY2027.

Combined supplier TAM by fiscal year, \$B; stacked by program. Nominal then-year, FY2022–FY2027.



Submarine procurement drives the cadence peaks.

- FY2024 and FY2027 reach \$5.82B and \$5.85B, respectively, when Virginia and Columbia Basic Construction fund together.

FY2026 is the DDG AP/LLTM exception.

- The combined year reaches \$4.83B; DDG peaks at ~\$1.23B, driven mostly by long-lead material ahead of construction.

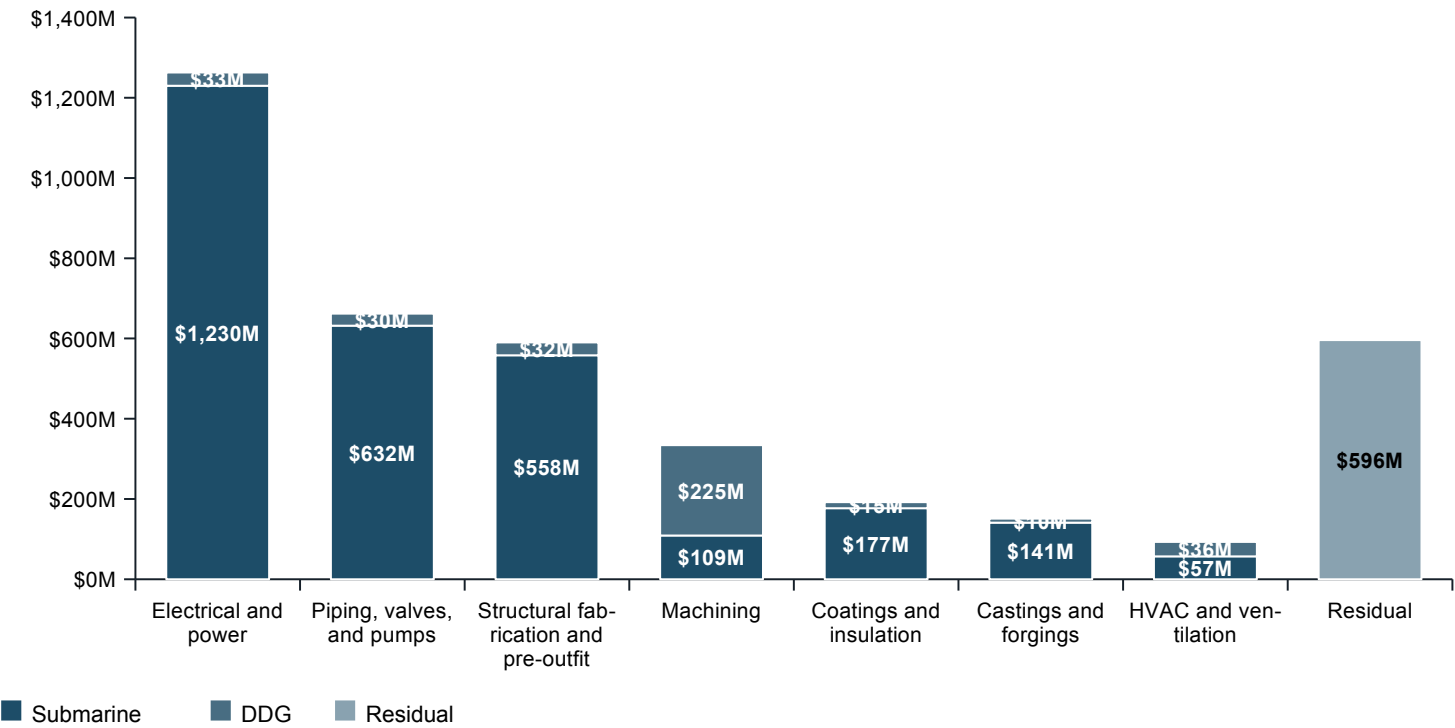
The \$3.88B average is a sizing convention.

- The six-year average smooths a path from \$1.91B in FY2022 to \$5.85B in FY2027; the annual shape reflects procurement cadence, not a steady run-rate.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) Virginia and Columbia annual procurement profiles; (3) DoD contract announcements and FFATA/FSRS subaward records

Work-Type Mix | Electrical power, piping, and structural fabrication lead the combined pool; a \$596M residual stays separate.

Combined average annual TAM by work type, \$M; stacked by program. Residual shown separate. Nominal then-year.



Three buckets dominate the named supplier pool.

- Electrical power, piping, and structural fabrication total ~\$2.52B of the \$3.88B TAM, led by \$1,263M of electrical and power work.

Entity-resolved classification shifts the mix, not the TAM headline.

- Machining rises to \$334M once VLS launcher-cell machining and LM2500 propulsion machinery are assigned to physical work type; combat electronics leave the supplier base.

Residual remains visible outside named-bucket SAM.

- \$596M of supplier dollars lacks recoverable bucket evidence, so it stays in TAM while remaining outside broad SAM.

The component pool is not one thing; entry, qualification, and incumbent dynamics differ sharply by work type.

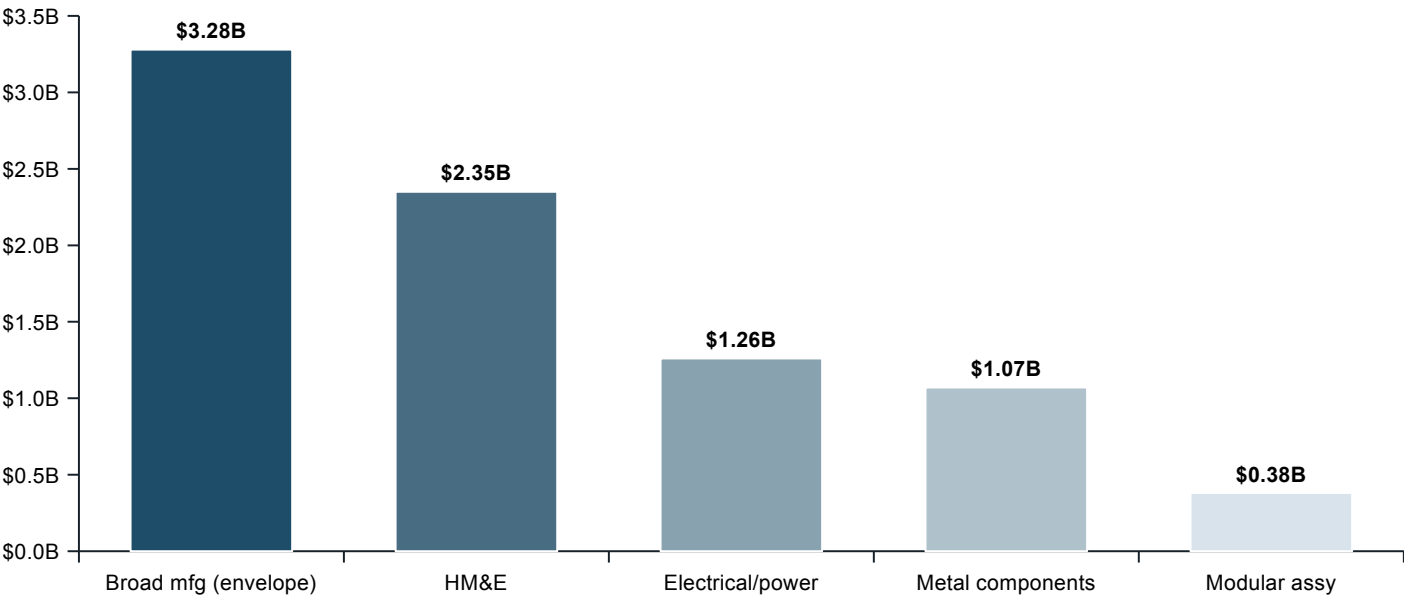


Sources: (1) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026; (2) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (3) parent-prime PIID program-scope attribution



SAM Scenarios | Broad component manufacturing is the \$3.28B envelope; HM&E is the largest cut at \$2.35B and modular is the smallest at \$381M.

Average annual scenario SAM, \$B, FY2022–FY2027; nominal then-year. Each scenario is an inclusion cut of one TAM, not additive.



Scenario composition **Broad** = all seven buckets; **HM&E** = machining, piping/valves/pumps, electrical/power, and HVAC; **Electrical** = electrical/power only; **Metal** = structural/pre-outfit, machining, and castings/forgings; **Modular** = entity-flagged module-assembly work, not a bucket union

Broad is the serviceable envelope, not another market.

- Broad component manufacturing is \$3.28B per year, equal to \$3.88B of TAM less the \$596M residual; every other scenario is a cut of that envelope.

HM&E is the largest overlapping cut.

- HM&E reaches \$2.35B because it includes electrical power, machining, piping/valves/pumps, and HVAC; electrical alone is \$1.26B and metal is \$1.07B.

Modular is smaller but most aligned with distributed build.

- The \$381M modular cut is entity-flagged module-assembly work, not a structural-and-coatings union.

Scenario SAM only. No SOM, share capture, win probability, or pricing haircut is modeled. Scenarios are overlapping cuts of one TAM; do not sum across them.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) FFATA/FSRS subaward records and the operating-entity supplier registry; (3) DDG and submarine SAM scenario inclusion-cut definitions

Evidence and Implications

Visible filings prove the layer but understate it; a narrow gated wedge is the realistic entry



Supplier Visibility | Visible suppliers prove the layer exists, but first-tier filings capture only a floor.

DDG VISIBLE SUPPLIERS

Leonardo

Arctic Slope

Major Tool

GE

Rolls-Royce

Northrop Grumman

Johnson Controls

SUBMARINE VISIBLE SUPPLIERS

Northrop Grumman

Leonardo

Curtiss-Wright Electro-Mechanical

Scot Forge

DC Fabricators

Rhoads

Graham

Austal

DDG FFATA-VISIBLE YARD-SIDE FLOW

~\$2.73B vs estimated yard-side outsourcing midpoint ~\$13.57B

SUBMARINE SUPPLIER-ADDRESSABLE VISIBLE VALUE

~\$5.46B across ~150 classified recipients; broader visible parent universe ~759

Reporting limits: reporting threshold and lag; yard filing differences; HII Newport News visibility gap

Supply-chain depth: direct material booked as direct cost; lower-tier subcontracting; standing agreements

Visible supplier names prove the layer exists, not the target list.

- DDG and submarine filings identify real suppliers and work-type cues, but some visible names are services, prime-affiliated, or outside the physical-component boundary.

First-tier filings set a floor, not the market size.

- Visibility thins below tier one and varies by yard, so TAM is sized from the budget base rather than by summing subawards.

Sources: (1) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026; (2) DDG MYP-corrected outside-yards place-of-performance evidence; (3) submarine visible-supplier classification and parent-universe analysis



Where to Play | Start diligence with modular assemblies and broad component manufacturing, then triage the rest by qualification burden and openness.

LANE	COMBINED VALUE (\$/YR)	DDG RELEVANCE	SUBMARINE RELEVANCE	STRATEGIC FIT	QUALIFICATION BURDEN	EVIDENCE CONFIDENCE	PRIORITY READ
Modular assemblies	\$381M per year	High	Medium	High	Medium	Medium	Start here; validate open work and qualification path
Broad component manufacturing	\$3.28B per year	High	High	Medium	Mixed	High	Useful envelope, too broad without bucket-level triage
Structural fabrication and pre-outfit	\$590M per year	High	High	High	Medium	Medium	Closest physical fit to distributed structural work
Electrical and power	\$1.26B per year	Medium	High	Medium	High	Medium	Largest bucket, but likely incumbent- and qualification-heavy
Piping and HM&E	\$2.35B HM&E scenario	Medium	High	Medium	High	Medium	Attractive, but product-by-product qualification matters
Machining, castings, and forgings	\$485M per year	Medium	Medium	Medium	Medium	Low	Smaller dollars, but potential bottleneck areas
Coatings, insulation, and HVAC	\$285M per year	Low	Medium	Low	Medium	Low	Module-attach or specialized materials work

The scorecard orders diligence, not expected wins.

- The table points to where to look first; the decision still depends on fit, qualification path, incumbent lock-up, and whether work can move to a new site.

Modular leads because it best matches the entry thesis.

- The modular cut is only \$381M, but it is most aligned with distributed build; broad component manufacturing remains the sizing envelope to triage by bucket.

Sources: (1) modeled scenario SAM, FY2022–FY2027; (2) FFATA/FSRS supplier-visibility evidence and qualification-burden assessment; (3) DDG and submarine program implications analysis



Entry Wedge | Modular and component work may be an entry wedge, but backlog, award lock-up, and margin require diligence before any conclusion.

Modular structural units, pre-outfit, machining, and broad component manufacturing may reach revenue earlier than prime-level ship construction.

Demand and backlog

Test backlog depth by bucket; prime need for external capacity; near-term procurement timing; open and recompete work.

Have policy and prime direction of travel; modeled TAM and SAM; visible supplier layer.

Need supplier-level backlog; award status; delivery schedule; bottleneck items.

Entry feasibility

Test qualification path; QA and Navy standards; facility and capex; weld, cyber, and certification; prime onboarding.

Have work-type buckets and supplier examples; modular and structural alignment.

Need exact qualification timelines; make-buy posture by prime; part-family eligibility.

Economics and margin

Test prime versus supplier margin pools; pricing power in bottlenecked parts; learning curve, labor, and subcontract-tier economics.

Have prime-level margin context and visible public supplier flow.

Need third-tier economics; comparable supplier margins; contract type; cost-plus versus fixed-price exposure.

ENTRY LANE	SPEED TO REVENUE	QUALIFICATION BURDEN	CAPEX INTENSITY	INCUMBENT LOCK-UP	EVIDENCE CONFIDENCE
Structural/pre-outfit	High	Medium	Medium	Medium	Medium
Coatings/insulation	High	Low	Low	Low	Low
Machining	Medium	Medium	Medium	Medium	Low
Piping/valves/pumps	Medium	High	Medium	Medium	Medium
HVAC	Medium	Medium	Low	Medium	Low
Electrical/power	Low	High	Medium	High	Medium
Castings/forgings	Low	High	High	High	Low

Due-diligence workplan

- Build a target-account list by bucket
 - Map open awards and recompete windows
 - Interview primes and tier-1 suppliers on make-buy posture
 - Validate qualification path and capex by part family
 - Test margin pool using public comps and supplier interviews
- The entry-wedge case still needs proof.**
- Backlog depth, open awards, make-buy posture, qualification path, and part-family economics determine whether lower-barrier work is investable.
- Openness should lead the screen before size.**
- Modeled SAM defines the pool to test; the first diligence question is which parts are open, qualifiable, and economically attractive.

Separate market size from investability before drawing a conclusion.

Sources: (1) DDG and submarine program implications analysis; (2) modeled TAM and SAM by work-type bucket, FY2022–FY2027; (3) FFATA/FSRS supplier-visibility evidence and stated opportunity hypotheses

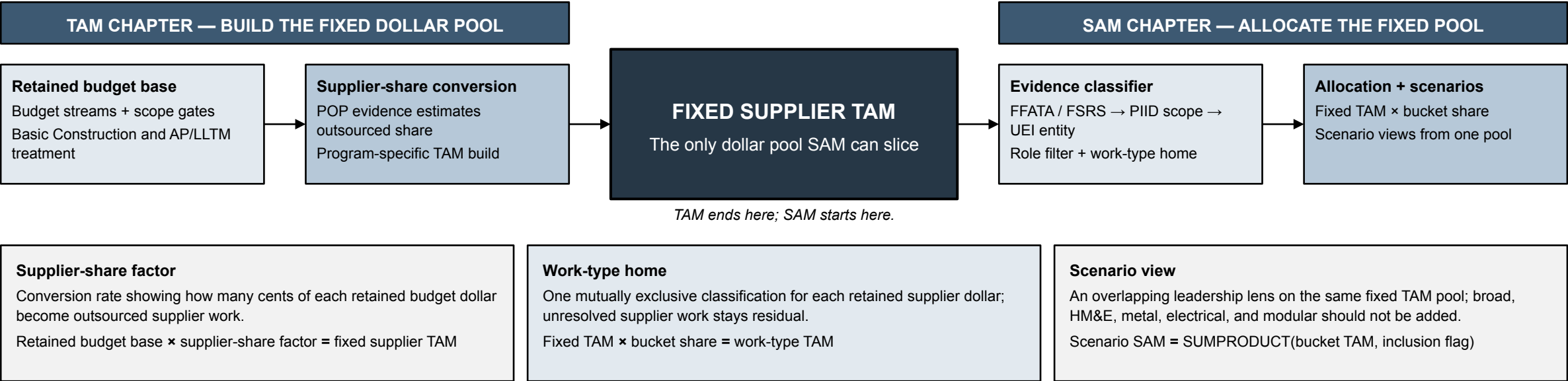


Appendix



Methodology Roadmap | TAM creates the fixed supplier pool; SAM slices it without resizing the market.

TAM sizing window FY2022–FY2027; SAM evidence window FY2017–FY2026. Budget and POP evidence size the dollar pool; award evidence allocates the pool.



TAM and SAM answer different questions.

- TAM sizes how much outsourced supplier-addressable new-construction work exists; SAM identifies which work-type slices are serviceable inside that fixed pool.

Award evidence moves allocation, not headline size.

- PIID, UEI, registry, role filters, residual treatment, and scenario flags can move bucket shares and scenario SAM, but not budget-derived TAM.

Guardrails: POP evidence creates supplier-share factors, not a summed award numerator. FFATA / FSRS evidence creates mix percentages, not market size. Scenario views overlap and should not be added. No SOM, capture rate, win probability, pricing haircut, or revenue conversion is modeled.

Sources: (1) Navy SCN budget justification books, FY2022–FY2027; (2) DoD/DoW contract-announcement POP corpus, CY2022–CY2026; (3) SAM.gov FSRS / FFATA subawards and SAM.gov / USAspending UEI / NAICS enrichment, FY2017–FY2026



TAM Budget Base | Scope gates define the retained base before supplier-share conversion.

FY2022–FY2027 nominal then-year dollars. DDG and submarine are run separately; this page stops at the retained budget stream bases sent to the supplier-share step.

Retained-base output: retained stream bases by program: DDG Basic Construction + additive AP/LLTM base; submarine Basic Construction + \$0 additive AP/LLTM base. Supplier-share factors are applied next.

Retained-base ledger: what survives into supplier-share conversion

STEP	SOURCE FIELDS USED	RULE APPLIED	RETAINED OUTPUT
1. Program budget base	Program; FY; appropriation; SCN / P-5c line; P-10 AP / LLTM / EOQ fields	Build the budget base by program and stream	Budget is the sizing base; award flow is not
2. Scope & cost boundary	Total ship estimate; GFE / GFP; ordnance; plans; nuclear; sustainment / depot; design-only	Retain the new-construction component boundary; remove non-component scope	Total ship cost stays context
3. Basic Construction stream	P-5c Basic Construction; program flag; FY flag; non-GFE construction base	Treat Basic Construction as the core retained denominator	BC retained base by program
4. AP/LLTM incrementality	AP / LLTM / EOQ amount; overlap & incrementality flags; DDG construction share; submarine AP-bridge exclusions	Add only where incremental to Basic Construction	DDG additive; submarine nets to \$0
5. Handoff to next step	Retained BC base; retained AP/LLTM base; stream flag; program flag	Send the bases to supplier-share conversion	No blended supplier-share factors; no supplier TAM yet

PROGRAM TREATMENT: RUN SEPARATELY
DDG
Core base: Basic Construction, LI 2122. AP/LLTM: additive, ship-construction-share filtered. Output: BC base + additive AP/LLTM base.
SUBMARINE
Core base: Virginia + Columbia Basic Construction. AP/LLTM: reference-only; additive base = \$0. Output: BC base + \$0 additive AP/LLTM base.

- Retained base stays program-specific.**
- DDG and submarine stay separate because AP/LLTM rules and supplier-share evidence differ.
- Conversion happens next.**
- This page sizes eligible retained budget dollars; the next page applies supplier-share factors.

AP/LLTM incrementality test: add only when not already inside the Basic Construction boundary

CY AP in window \$1.8B	✖	Ship-construction share 80%	=	Retained AP/LLTM base \$1.5B	Gross P-10 AP \$44.7B	–	GFE / design / weapons \$27.3B	–	Already in BC \$16.8B	–	Overlap \$0.6B	=	Additive AP base \$0
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The next step applies the AP/LLTM supplier-share factor; this page stops at the retained base.

AP/LLTM stays useful evidence for supplier-share context, but it is not an incremental base.

Sources: (1) Navy SCN budget justification books, FY2022–FY2027; (2) P-5c Basic Construction and P-10 AP/LLTM budget exhibits; (3) DDG-51, Virginia, and Columbia scope-exclusion reconciliations

Supplier-Share Method | POP evidence converts retained budget into fixed outsourced supplier TAM.

FY2022–FY2027 retained budget bases from the budget-base step. POP records estimate supplier share; they are not summed as the TAM numerator.

Supplier-share factor = conversion rate that turns retained budget dollars into outsourced supplier TAM. It answers: *for every \$1 of retained budget base, how many cents become supplier work?*

How POP evidence becomes the supplier-share factor

Eligible POP dollars
Confirmed gated in-scope

×

Supplier-location share
Other US + Foreign POP %

=

Weighted supplier dollars
Dollar-weighted footprint

+

Eligible POP dollars
Same gated denominator

=

Supplier-share factor
Applied to retained base

DDG footprint: *BIW % + Ingalls % excluded; Other US % + Foreign % counted as supplier share.*

DDG: ADDITIVE AP/LLTM STREAM			
Basic Construction base \$17.5B	×	BC supplier-share factor 13%	=
			BC supplier TAM \$2.2B
			+
Retained AP/LLTM base \$1.5B	×	AP/LLTM supplier factor 85%	=
			AP/LLTM supplier TAM \$1.3B
DDG supplier TAM \$3.4B cumulative / \$573M per year			
154 modeled POP actions	\$21.7B gated corpus	\$6.1B excluded / GFE	\$14.6B MYP restored

Submarine footprint: *EB % + HII % excluded; Other US % + Foreign % counted as supplier share.*

SUBMARINE: AP/LLTM REFERENCE-ONLY						
Basic Construction base \$56.7B		×	BC supplier-share factor 35%		=	BC supplier TAM \$19.8B
AP/LLTM additive base \$0		×	Reference factor 48%		=	Additive TAM \$0
AP/LLTM is supplier evidence, not incremental TAM: already inside BC or out of scope.						
Submarine supplier TAM \$19.8B cumulative / \$3.3B per year						
658 POP rows		\$25.4B gated corpus			\$6.1B excluded / GFE	

FIXED OUTSOURCED SUPPLIER TAM

\$23.3B cumulative / \$3.9B per year

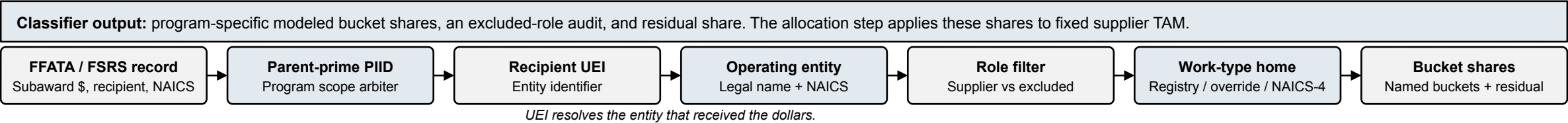
This is the fixed dollar pool the allocation step will slice; SAM does not resize it.

- The supplier-share factor is a conversion rate, not an award total.**
- POP records estimate the supplier share of retained budget dollars; the Navy budget base still supplies the TAM dollars.
- DDG and submarine stay separate until outputs are summed.**
- The two programs have different yard footprints and AP/LLTM treatment, so the model sums TAM dollars rather than blending supplier-share factors.

Sources: (1) DoD/DoW contract-announcement POP corpus, CY2022–CY2026; (2) FPDS Atom feeds and DDG MYP master-total reconstruction; (3) Navy SCN P-5c / P-10 budget exhibits, FY2022–FY2027

SAM Classifier | PIID scopes the award, UEI resolves the operating entity, and the registry assigns one work-type home.

FY2017–FY2026 FFATA / FSRS evidence window. Award records set supplier mix percentages; the fixed TAM dollar pool comes from the supplier-share step.



Classification ledger: how a record becomes one work-type home

STAGE	SOURCE FIELDS USED	RULE APPLIED	OUTPUT / GUARDRAIL
1. Award pull	Subaward ID, parent PIID, UEI, NAICS, \$	Collect visible supplier evidence	Mix, not TAM
2. PIID scope	Parent-prime + discovered PIID flags	Scope to DDG or submarine	Not hull-level proof
3. Entity resolution	sub_entity_uei, legal name, NAICS	Resolve the operating entity paid	Not parent brand
4. Role filter	Role, prime / co-prime, GFE, FMS flags	Drop non-supplier / non-component	Excluded roles leave base
5. Work-type assign	Registry bucket, override, NAICS-4	First clean match wins one home	QA, not classifier
6. Residual discipline	Addressable, bucketed, unbucketed \$	Keep unresolved \$ in denominator	Dilutes named shares
7. Share output	Bucket \$ over addressable base	Modeled work-type and residual %	Feeds allocation step

CLASSIFICATION PRECEDENCE: FIRST CLEAN MATCH WINS

1. Registry / operating-entity UEI assigns role and bucket.

2. Prime / co-prime name excludes final-assembly yards.

3. GFE / SIB / Navy-directed name routes out of supplier SAM.

4. Vendor-name override handles known exceptions.

5. NAICS-4 fallback assigns a bucket where registry is missing.

6. Service / non-component NAICS routes to excluded service.

7. Holding-company NAICS 5511 routes to excluded holding.

8. Foreign / FMS routes out of U.S. supplier SAM.

9. Residual stays supplier / unbucketed when unresolved.

Work-type homes: one per retained supplier dollar

Structural / modules

Machining / propulsion

Castings / forgings

Piping / fluid handling

Electrical power

HVAC / chilled water

Coatings / insulation

Residual / unbucketed

The classifier is entity-first and role-first.

- PIID scopes; UEI resolves the entity; roles filtered first.

Residual is a discipline, not a failure.

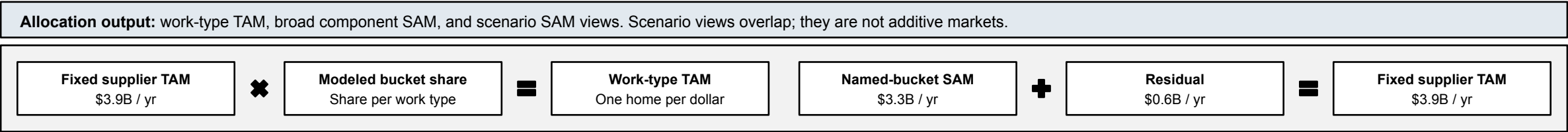
- Unresolved dollars stay in the denominator, outside SAM.

DDG EVIDENCE BASE	SUBMARINE EVIDENCE BASE	Examples: why operating entity beats parent brand
22,867 FFATA / FSRS records	929 classified entity rows	Northrop Sunnysvale generators
118 discovered PIIDs	\$6.1B all-recipient base	Curtiss-Wright / Circor valves
1,555 cleaned vendors	\$5.5B supplier-addressable	Lockheed / Raytheon Aegis, SPY-6
151 NAICS lookups	\$664.8M unbucketed value	Blank NAICS, no registry match
\$6.0B supplier-addressable	12% residual / unbucketed	

Sources: (1) SAM.gov FSRS / FFATA first-tier subaward records, FY2017–FY2026; (2) SAM.gov Entity API and USAspending UEI / NAICS enrichment; (3) FPDS / DoD PIID records and program scope-exclusion registers

SAM Allocation | Fixed TAM is sliced into work-type homes and overlapping scenario views.

FY2022–FY2027 average annual TAM and SAM; nominal then-year. The supplier-share step supplies the fixed TAM dollar pool; the evidence classifier supplies modeled work-type shares and residual.



Work-type allocation: mutually exclusive homes mapped into overlapping scenario views

WORK-TYPE HOME	AVG TAM (\$M)	BROAD	HM&E	METAL	ELECTRICAL
Electrical power / distribution / generation	\$1,263M	Yes	Yes	No	Yes
Piping / valves / pumps	\$662M	Yes	Yes	No	No
Structural fabrication / modules	\$590M	Yes	No	Yes	No
Machining / mechanical / propulsion	\$334M	Yes	Yes	Yes	No
Coatings / insulation / decking	\$192M	Yes	No	No	No
Castings / forgings	\$151M	Yes	No	Yes	No
HVAC / ventilation / chilled water	\$93M	Yes	Yes	No	No
Residual / unbucketed	\$596M	No	No	No	No

Residual treatment: residual stays inside fixed TAM and dilutes named-bucket shares, but stays outside broad and targeted scenario SAM.

Modular treatment: modular assemblies come from registry entity flags, not from summing one or more work-type rows.

SCENARIO VIEWS: CUTS OF ONE TAM POOL

BROAD COMPONENT MFG
\$3.3B / yr
\$19.7B cumulative; all seven named work-type homes

HM&E
\$2.4B / yr
machining + piping + electrical + HVAC

ELECTRICAL / POWER
\$1.3B / yr
ship power, distribution, generation only

METAL COMPONENTS
\$1.1B / yr
structural + machining + castings

MODULAR ASSEMBLIES
\$381M / yr
entity-flagged suppliers, not bucket union

Do not add scenarios: broad, HM&E, electrical, metal, and modular are overlapping leadership views of one fixed supplier TAM pool.

Work types classify dollars once; scenarios re-cut them multiple ways.

- Each supplier dollar is assigned one work-type home or residual, but Broad, HM&E, Metal, Electrical, and Modular are overlapping views built from those homes and flags.

Residual keeps the allocation conservative.

- Unresolved supplier dollars remain in the fixed TAM reconciliation, but they stay outside named scenario SAM until registry, NAICS, or override evidence supports assignment.

Sources: (1) SAM.gov FSRS / FFATA subaward records, FY2017–FY2026; (2) SAM.gov Entity API / USAspending NAICS enrichment and work-type evidence; (3) DoD/DoW POP corpus and Navy SCN budget books supporting fixed supplier TAM